

Advanced Animal Movement Modelling Training Workshop

Bayesian Modelling for Ecological Inference

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What is Bayesian statistics?

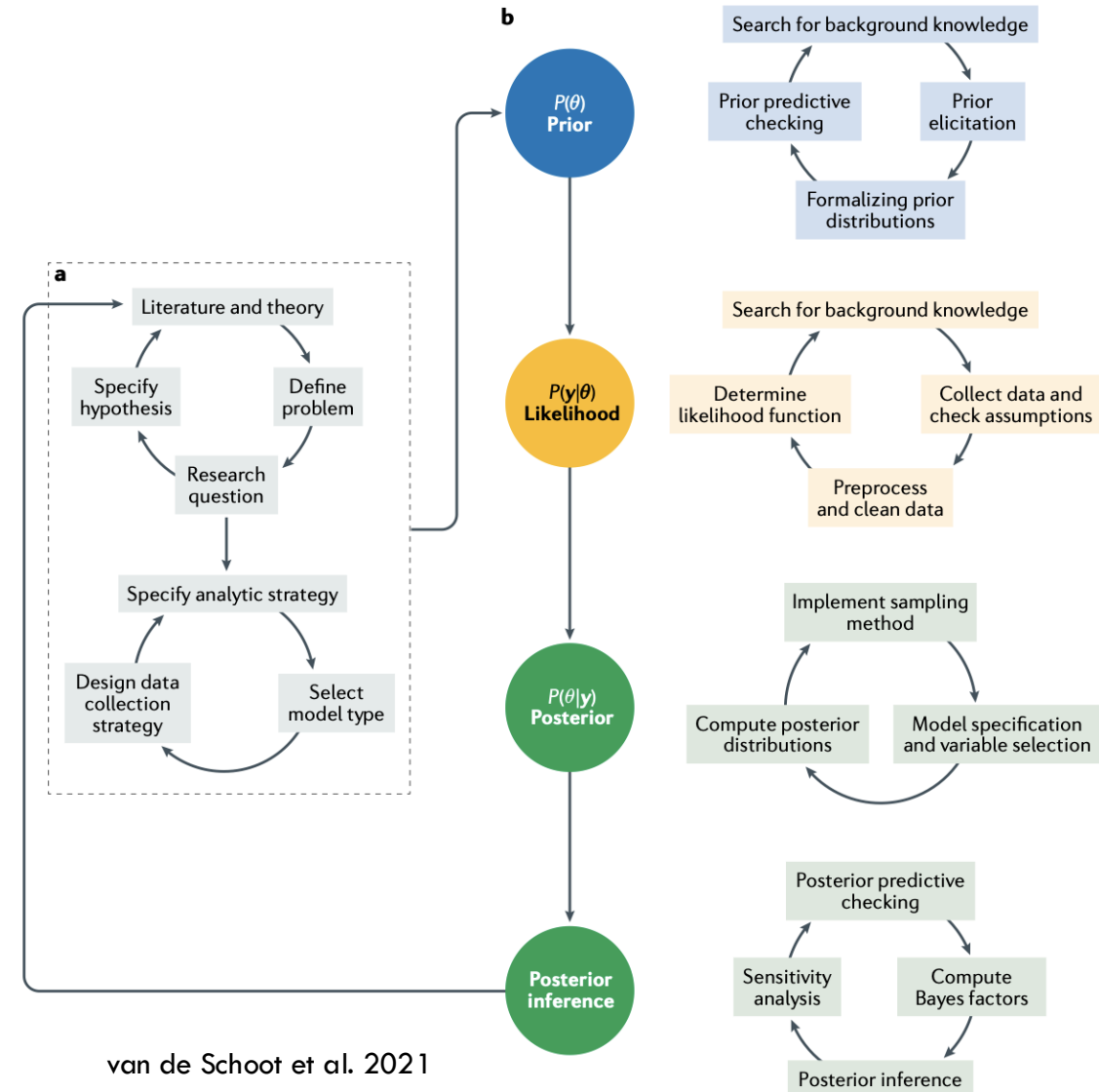
At its simplest, follows Bayes' Rule:

- $$p(\theta|\mathbf{y}) = \frac{p(\mathbf{y}|\theta)p(\theta)}{p(\mathbf{y})},$$

where $\mathbf{y}$ represents the data and θ represents the model parameters.

Relies on use of conditional probabilities

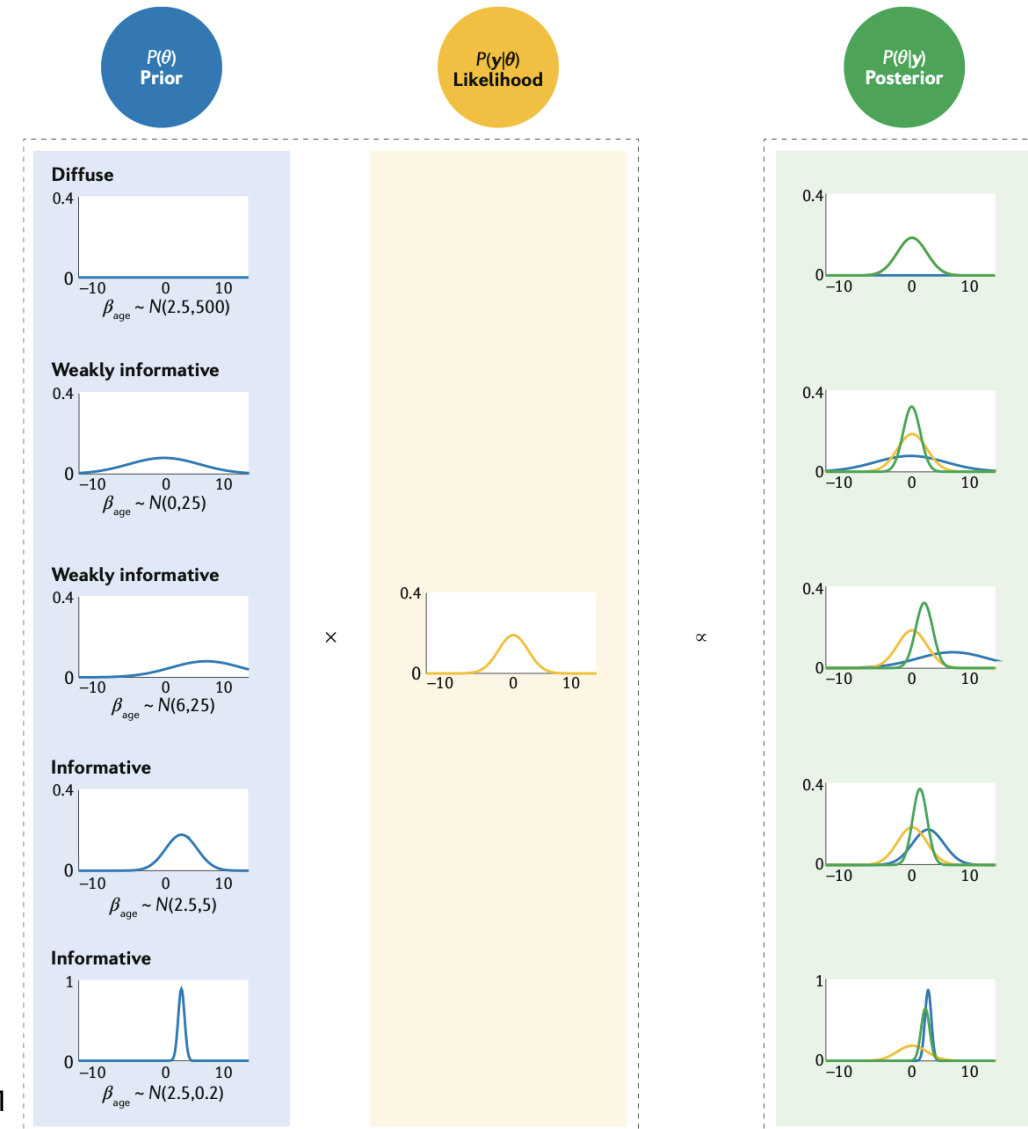
- e.g., $p(\theta|\mathbf{y})$ represents the probability of parameters given (|) the data



What is Bayesian statistics?

Posterior Belief = Likelihood of Data x Prior Info

- **Prior:** What we know before seeing data (domain knowledge, physical limits)
- **Likelihood:** How probable observed data are under possible parameter values
- **Posterior:** The updated probability distribution of parameters after combining prior and data



Frequentist vs. Bayesian statistics

Feature	Frequentist Model	Bayesian Model
Parameters	Fixed, unknown true values	Random variables with probability distributions
Main Output	Point estimates with some measure of uncertainty (e.g., SE)	Full posterior probability distribution
Uncertainty	Confidence interval	Credible interval
Prior Knowledge	Can include penalties and regularization, but typically assumed to be uninformative	Explicitly incorporates prior domain knowledge or constraints
Small N	Can suffer from overfitting, singular fits, or asymptotic failure	Regularizing priors stabilize estimates and prevent extreme overestimation

Why Bayesian models excel with small N

Regularizing Priors Prevent Overfitting

- Weakly informative priors can help with estimation of biologically plausible parameters

Partial Pooling in Hierarchical Models

- Groups with small sample size have parameter values pulled towards the population mean until data suggest otherwise

No Reliance on Asymptotic Assumptions

- Frequentist p -values and confidence intervals assume large N for central limit theorem to hold. Bayesian models estimate exact posterior distributions regardless of N

Avoids Singular Fit Errors

- Frequentist random-effects algorithms often crash or output zero variance estimates (singular fits) when group numbers are small (e.g., $N < 5$). Bayesian models stabilize group variance estimates via priors

The Bayesian engine: MCMC sampling

Markov Chain Monte Carlo (MCMC) draws samples from posterior

- Whose distribution is typically unknown (unless using conjugate likelihood and prior distributions)

Approach	Short description
<i>MCMC-based methods</i>	
Metropolis–Hastings	An algorithm used for obtaining random samples from a probability distribution. Uses a general proposal distribution, with an associated accept/reject step for the proposed parameter value(s) ^{85,86}
Reversible jump MCMC	An extension of the Metropolis–Hastings algorithm. Permits simulation of trans-dimensional moves within parameter space ^{35,225}
Hamiltonian Monte Carlo	A Metropolis–Hastings algorithm based on Hamiltonian dynamics ⁸⁷ . This algorithm is useful if direct sampling is difficult, if the sample size is small or when autocorrelation is high. The algorithm avoids the random walk of Metropolis–Hastings and sensitivity by taking a series of steps informed by first-order gradient information. The No-U-Turn Sampler ²²⁶ is an extension and is often faster because it often avoids the need for tuning the model
Gibbs sampler	A Metropolis–Hastings algorithm where the proposal distribution is the corresponding posterior conditional distribution, with an associated acceptance probability of 1 (REF. ⁸⁴)
Particle MCMC	A combined sequential Monte Carlo algorithm and MCMC used when the likelihood is analytically intractable ¹⁷⁷
Evolutionary Monte Carlo	An MCMC algorithm that incorporates features of genetic algorithms and simulated annealing ²²⁷ . It allows the Markov chain to effectively and efficiently explore the parameter space and avoid getting trapped at local modes of the posterior distribution. It is particularly useful when the target distribution function is high-dimensional or multimodal

Alternatives to MCMC

Often use some form of approximation, which tends to be quicker to fit in practice

<i>Non-MCMC-based methods</i>	
Sequential Monte Carlo	An algorithm based on multiple importance sampling steps for each observed data point. Often used for online or real-time processing of data arrivals ²²⁸
Approximate Bayesian computation	An approximate approach, typically used when the likelihood function is analytically intractable or very computationally expensive ²²⁹
Integrated nested Laplace approximations	An approximate approach developed for the large class of latent Gaussian models, which includes generalized additive spline models, Gaussian Markov processes and random fields ²³⁰
Variational Bayes	Variational inference describes a technique to approximate posterior distributions via simpler approximating distributions. The popular mean-field approximation assigns an approximating variational distribution to each parameter independently. Gradient descent is then used to optimize the variational parameters to minimize a loss function known as the evidence lower bound ⁹⁹

Software for fitting Bayesian models

Possible to code your own custom sampler, but many options available that are easier to use and assess model performance/diagnostics/convergence

Software package	Summary	Specialized software doing Bayesian inference for particular classes of models	
General-purpose Bayesian inference software		JASP ²⁴⁵	A user-friendly, higher-level interface offering Bayesian analysis. Open source and relies on a collection of open-source R packages
BUGS ^{231,232}	The original general-purpose Bayesian inference engine, in different incarnations. These use Gibbs and Metropolis sampling. Windows-based software (WinBUGS ²³³) with a user-specified model and a black-box MCMC algorithm. Developments include an open-source version (OpenBUGS ²³⁴) also available on Linux and Mac	R-INLA ²³⁰	An open-source R package for implementing INLA ²⁴⁶ . Fast inference in R for a certain set of hierarchical models using nested Laplace approximations
JAGS ²³⁵	An open-source variation of BUGS that can run cross-platform and can run from R via rjags ²³⁶	GPstuff ²⁴⁷	Fast approximate Bayesian inference for Gaussian processes using expectation propagation; runs in MATLAB, Octave and R
PyMC3 ²³⁷	An open-source framework for Bayesian modelling and inference entirely within Python; includes Gibbs sampling and Hamiltonian Monte Carlo		
Stan ⁹⁸	An open-source, general-purpose Bayesian inference engine using Hamiltonian Monte Carlo; can be run from R, Python, Julia, MATLAB and Stata		
NIMBLE ²³⁸	Generalization of the BUGS language in R; includes sequential Monte Carlo as well as MCMC. Open-source R package using BUGS/JAGS-model language to develop a model; different algorithms for model fitting including MCMC and sequential Monte Carlo approaches. Includes the ability to write novel algorithms		

Checking for convergence on posterior

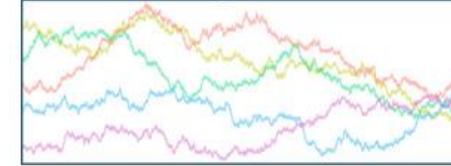
- Need to make sure model sampled correctly and thoroughly from posterior
- Diagnostics for assessing convergence typically include:
 1. Visual check of traceplots
 2. Inspection of $\hat{R}$ values
 3. Inspection of effective sample size for posterior draws

MCMC trace plot diagnostics

Chains converged and mixing well



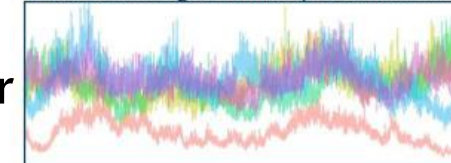
Chains not converged



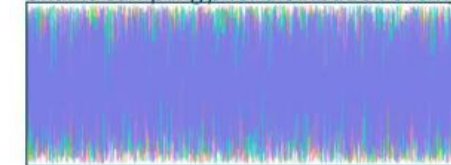
Chains converged, not mixing well



Chains converged to multiple solutions



Chains sampling prior distribution evenly



Action

Make inference

Run MCMC for longer

Try alternative
MCMC algorithms

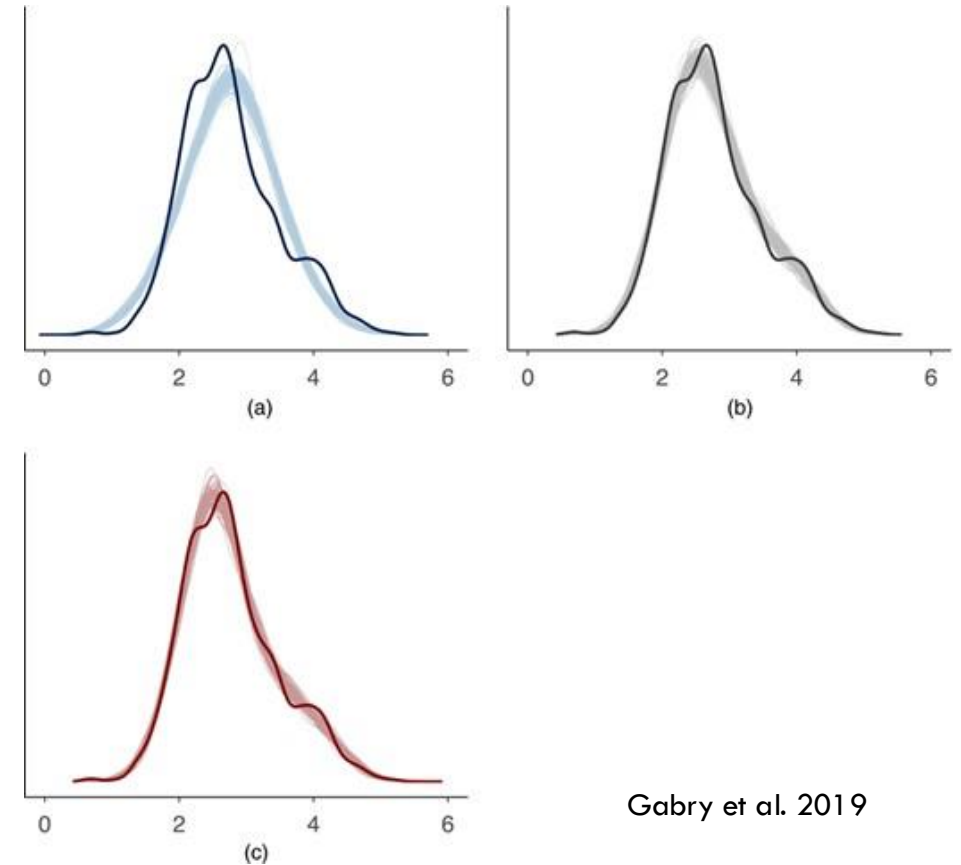
Re-evaluate model structure

Check identifiability

Check prior sensitivity

Goodness-of-fit via posterior predictive checks

- Unlike standard tests used for frequentist models (e.g., QQ plots, plots of residuals), we predict from the model to check how close it matches the data
- Qualitative step to determine whether the model fits the data well
 - Different from convergence check since the model could converge but still be poor



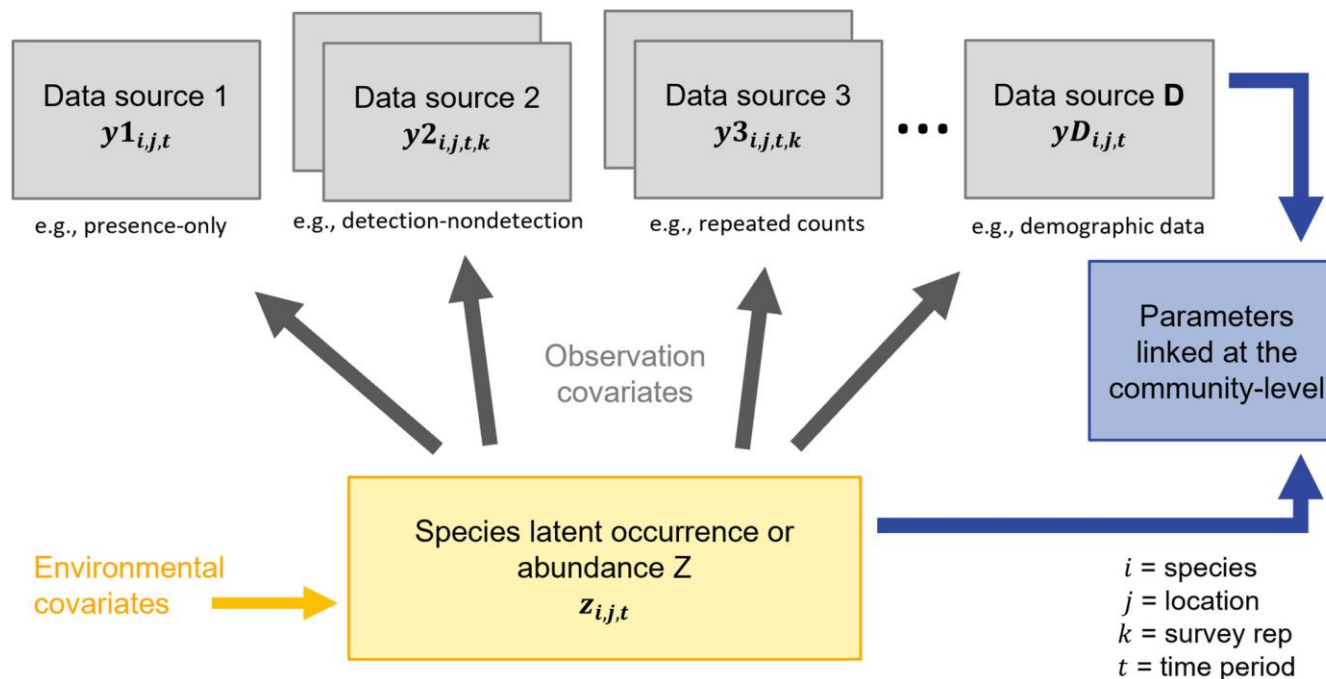
Gabry et al. 2019

Matching data types to distributions

Data property	Distributions	Data examples
Positive, continuous, right-skewed	Log-normal, Gamma	Home range area, step lengths, daily distance traveled
Counts/integers	Poisson, Negative Binomial, Tweedie	Abundance, visit frequency, species richness
Binary	Bernoulli, Binomial	Survival, occupancy
Proportion	Beta	Proportion behavior, proportion land cover

Bayesian models provide unlimited customizability

- Model parameters beyond mean values (e.g., predict variability in variance) and combine data in different ways
- Beyond GLMs, Bayesian models can also be used to construct SSMs/HMMs and other mechanistic models



When to choose Bayesian vs. Frequentist models

Prefer Frequentist:

- for quickly modeling massive datasets
- when fitting standard and/or simple models
- when it's not clear how to define the priors (e.g., Bayesian GAMs)

Prefer Bayesian:

- for small N
- for estimating probabilities
- for fully propagating uncertainty
- when designing complex, custom models
- when wanting a full posterior distribution for inference

Want to learn more about Bayesian stats?

- Watch Richard McElreath's great set of lectures on YouTube based on his book (Statistical Rethinking 2nd Ed.)!
- Bayesian Data Analysis (3rd Ed.) by Andrew Gelman et al. available as a free PDF online
- An Introduction to Bayesian Thinking by Merlise Clyde et al. and available as an e-book

Time for live-coding



Fit example Bayesian workflow, fit hierarchical RSF/SSF
w/ INLA, add spatial random effect to RSF